

Prof. Karl Friston ~ Applied Active Inference Symposium pt. 2 (Communication)

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Hello, welcome back. This is the second session of the ACT-INF Lab Symposium on June 21st, 2021, and we're here in the .coms, or Communication Organizational Unit. The goal of .coms is to organize the lab's internal projects and activities, as well as to carry out all forms of communication with external entities. So it's like our connective tissue and our neuroectoderm, in a way. What has been done so far is that we've had about 75 live streams, variously on presentations and participatory discussions, since July 28th, 2020. And we're taking an active inference approach to communication and learning by doing. Our very first live stream was on the paper, Narrative as Active Inference, and shortly after, A World Unto Itself, Human Communication as Active Inference. So that's something that we like thinking about here and that we want to explore.

And also, some of our lab members framed online communication and team collaboration in terms of active inference in the 2020 paper, Active Inference and Behavior Engineering for Teams.

Our aim here is to make active inference accessible, well-known, and well-understood.

So let's get right to the questions, Carl. The first question is, how can we show, not tell, the idea of active inference? For example, through embodied experience, experiments, or what other mechanisms? How can we best communicate in a way that makes it resonate?

Well, pursuing that very interesting notion and using the principles of active inference to optimize the role of the dot-coms team, then at its simplest, active inference means that the imperatives for all behavior, and it's likely that most behavior is of an epistemic sort, is to resolve uncertainty. So if you want to engage people and be a service to the people you want to engage, which may be

internal members of an internal member of your own team, then you've got to know what they don't know. Because that will define the epistemic affordances that will get them engaged with you and you with them. that will incur the best kind of belief updating.

So, you know, practically what that might mean is that it may well be that you, one has to identify didactic or informative illustrations that are tailored or specialized to the person or people that you're talking about.

So, you know, I have never thought about using embodied experiences before, but that's a brilliant idea. You're just illustrating to people who want to know how my body and my mind works to illustrate to them the mechanics of it working.

Using the language of active inference can be very, very powerful.

So as I'm talking, you know, one example of this would be using saccadic suppression as an illustration of the potency of getting your beliefs about the predictability of sensory evidence right.

So if you're a physiologist, you would know this is sensory attenuation.

If you were in machine learning, you're working with transformers, this would be, I think, attentional selection. Basically deploying the gain or switching on the right channels in order to select those data that are going to resolve the most uncertainty or maximize the information gain subsequent on this sort of covert action.

So that mental action is, you know, often sold as sort of mental action in the philosophy literature.

So that mental action is really endemic and a vital part of our sensory engagement with the world and beautifully illustrated by saccadic suppression.

So this is the, speaks to the notion of attenuating the sensory consequences of your own action so that any evidence that you're not actually acting is precluded from your belief updating.

So a clinical example of this would be Parkinson's disease, for example.

If I'm sitting still and I wasn't able to ignore all the messages from my muscles that tell me at the moment I am sitting still, then I am never going to be able to realize in a priori intention that I'm going to initiate a movement.

Because as soon as soon as I initiate a movement, I have, I put in place a plausible hypothesis that I'm lifting, I'm going to stand up immediately, all the evidence at hand suggests that I am not standing up.

So I'm going to revise my belief.

No, I'm not standing up.

I'm not in the process of standing up and it becomes impossible to move.

So that would be an example of what would happen if you didn't have this capacity to attend to or to select or apply the principles of optimal Bayesian design in terms of selecting those data for your own belief updating.

But a really pragmatic and easy example of that is saccadic suppression.

When we do the simplest of movements, epistemic foraging, which is moving our eyes, making saccadic eye movements.

Because when we do that, we actually induce masses of visual information on the photoreceptors in the retina, sometimes referred to as retinal slips.

So when I look from the left to the right, there's a flood of information that I have caused.

And yet it is not useful information because it doesn't tell me anything I didn't already know.

So what the brain does, it suppresses that information by transiently suspending the precision.

If you are the Kalman gain, if you are taking a Kalman filter like perspective on predictive coding as what kind of variational filtering.

And that's really easy to demonstrate to an audience.

You know, just get them to either fixate on a central stimulus and then pay attention to something that's moving around or the converse.

They fixate on the thing that's moving around and then ask some questions about the central stimulus.

And with the right timing, it's a very potent illustration of beyond just gathering information, but actively selecting and triaging that information in accord with the principles of optimal basing design.

So I haven't thought about going beyond that, but I'm sure there are lots of lovely examples of embodied experiences that really do illustrate the active inference in action, as it were.

I'm just reminded, because active inference could be read as if you like, a 21st century version of idea motor theories, which were very popular in the 19th century.

And of course, that was demonstrated through embodied experiences in a very alluring way through hypnotism and the like.

So I can imagine somebody doing a 21st century version of hypnotism and all those wonderful Victorian illusions about the way you use your sense organs or deploy them actively.

But now in the service of just illustrating some basic phenomena under right under right active inference in terms of experiments or the classic ones that immediately come to mind that are really engaging are visual illusions.

So on one reading, all visual illusions are just ways of getting at your perceptual priors in the context of Bayesian inference.

If you can conjure a particular pattern of sensory information that you know was caused in one way, and yet you think your subject or your audience has sufficiently precise prior beliefs that it could only be caused in another way,

which is not the way you caused it, and then you let them experience that, and then you reveal how you actually generated those data,

then that's a very powerful way of demonstrating the innate priors, the sort of formal priors in terms of the connect term or the sparse coupling on a factor graph.

And again, it's part of the lived experience.

So I think visual illusions would be, and there are loads of beautiful illusions out there,

and all that one would have to do is to harness their beauty and allure and use them as a vehicle to give people insight into their own,

usually sub-personal priors about the way that the world is constructed.

And then, I mean, in my world, what you generally try and do is to actually put this in silico by just creating little in silico creatures.

Because you've now got active inference with this information geometry,

you now have the opportunity not just to simulate what these creatures do,

but what they perceive, because now you've got the quantitative estimates of their posterior beliefs.

So you can actually show a subject who's just experienced a visual illusion that this is perfectly Bayes optimal,

and indeed, when you write down this variational message passing scheme in this synthetic subject,

this synthetic person also expresses exactly the same illusions.

And this is Bayes optimal for this kind of work.

So you can leverage active inference in that sense.

And I'm deliberately referring back to the .edu discussions.

What active inference brings to the table?

Because it's got information about stuff out there in the numerics,

you can go a lot further than you can if you were doing, say, deep learning or machine learning,

because you've got this dual asset, you've got this information geometry at hand.

The state of your variational autoencoder actually means something in relation to a belief about what generated those data,

which you can create lovely little movies, you know,

showing what this simulation of you was actually experiencing.

So, you know, I'm interested.

Are there any other ways that you've thought about in terms of showing people?

Thanks for the answer.

It's like, look left, look right.

Now you're an active inference agent.

And as far as potential avenues for embodiments,

some of the work with Ryan Smith and others,

bringing people into the somatosensory dimensions

and their own priors and expectations about their body and about motion could be very powerful,

as well as auditory modalities.

And indeed, active inference is a framework by which we can think about how our perceptions are related to our inference and our action.

So in various domains, I think there'll be excellent experiments.

And it brings us to our next question.

You actually addressed several areas in your answer.

You addressed machine learning, as well as neuroscience, as well as just everyday lived experience.

So how can active inference engage in better dialogue with adjacent areas?

For example, machine learning, systems engineering, psychiatry, and neuroscience,

as well as any other fields that you think are relevant to?

Then, you know, the obvious answer here is in either academic or commercial collaboration.

And what might you, what would license that?

I think, you know, the simplest answer is that the free energy principle and its, if you like,

teleological chorally active inference is not there and was never intended to replace extant theories.

It was there to prove, to endorse them and to reveal the interrelationships between them.

So anything that's worked and survived into the 21st century has some veracity and a proven utility.

And therefore, it's just a question of reformulating or changing the words so that people can see immediately how their particular formulation relates to somebody else's formulation,

where both formulations are special cases of the most generic and simplest explanation,

which would, you know, from my point of view, would be the free energy principle

and active inference in the case of sentience.

So I think there's, you know, as an integrative framework,

you're very well positioned to say, look, you know,

can we understand the way that you think about this?

Can we now articulate this either using simulations or, you know, mathematical analysis?

Can we understand what you've been doing in this integrative framework?

And if we can, can we show how it relates to another discipline's formulation of this problem?

And sometimes you can get synergistic or added value from doing that.

So, you know, there must be loads of examples that you've written,

machine learning systems engineering, psychiatry, neuroscience here.

So machine learning, for example, you know,

how would active inference help machine learning?

So at the moment, you know, there seems to be two answers floating around.

And we've already sort of discussed a couple of the key,
a couple of these issues in depth.

And so machine learning commits to usually a normative approach to good behavior
that can be quantified by a loss or a value function.

But we've just said, well, if we now want these machines to learn to act,
then we have to go beyond state action value functions
and consider the belief-based calculus that is active inference,
which is all about the reduction of uncertainty.

So now you are in a position to say, well, look,
if you consider your objective functions as a part of a more generic objective function,
think what you might be able to get from this.

And of course, what you might get from this is a deep learning scheme
that actually can now go and solicit the right kind of data to optimize its own learning.

And, you know, people in Bayesian RL might argue, well, yeah, that's what we're doing,
you know, with a series of bright ideas and heuristics to try and augment classical value functions.

But, you know, you can say, well, OK, you've clearly put a lot of work into that.

But there is actually a simple objective function already out there that is provably appropriate
to describe systems that self-organize and maintain themselves.

That actually has what you want.

Why don't you try this, for example?

So that would be one example.

You have to tread carefully because, you know,
a lot of people have dedicated their lives to solving these problems
and they're very reluctant to change their rhetoric or see their contributions as a special case.
But in many instances, certainly from my perspective, mathematically,
they are special cases.

And sometimes if you catch the entrepreneurs, the innovators, the creative academics
at the right stage in their career before they have committed to a particular church
or ideology or calculus or group or company,
you can actually point them in the right direction and become extremely creative.

I'm not so sure about systems engineering, but certainly I always celebrate the expected free energy
as with just taking away various bits and pieces, various sources of uncertainty
as reducing to KL control.

And then what I say is what KL control is, what grown-up engineers use in the control theoretic setting.
So that would be another example.

So, you know, you could also, I don't know this because it's not my field,

but certainly in terms of introducing, say, a fault tolerance in control theoretic approaches in engineering where the fault tolerance required uncertainty about the operation of some external part, you could again motivate a more complete objective function that takes you beyond KL control and introduces the information gain into the mix. Because to get from the complete objective function or to KL control, you have to ignore uncertainty about the latent states that are, in the mapping from latent states of the plant you're controlling to the sensors or the observables. So, you know, you're moving from a partially observed Markov decision process, for example, to an observable one.

And then expected biology becomes KL control or risk sensitive control in economics. So you could say, well, look, you know, why don't you just augment the KL control and then put this extra term in.

And now what you've got is a kind of anticipatory fault tolerance in the sense that if there's uncertainty about latent causes, that's automatically resolved in the way that you go and switch on various sensors or switch off various sensors.

And there's a principle way of doing that.

I think to have any influence, you need to be able to show or provide proof of principle that this more integrated, more universal approach to normative approach to problems can offer speed ups or increased efficiency or do what the people actually in that field wanted to do.

So, for example, you've got to be able to show that active inference can outperform sort of vanilla deep learning by an order of magnitude, which is easy to do because, of course, most benchmarks in machine learning are actually inference problems.

So if you just recast it, it's actually quite a trivial thing to do just by saying, well, actually, what you've been dealing with is an inference problem, which looks a lot like a one shot learning from the point of view of somebody in machine learning.

But I think there will be some pressure to get people's attention to make yourself attractive in terms of you. They will now have some, first of all, find you interesting and also have the potential that they can place an epistemic trust in you.

You've got to you've got to sort of give them a clue and a cue as to why they should should engage with you. And very often there's there's a two way to way or two road exchange.

And so one simple example of that, which I see emerging in the field is the use of deep learning to amortize certain mappings when they can be amortized in active inference schemes to evaluate the expected for energy, for example, doing very deep tree searches.

So that's that's the kind of innovation you see in coming out of 20 year olds at the moment who haven't yet decided whether we're going to do deep learning or active inference

because they want to do both and do it very, very effectively.

So that's a nice example from my perspective of a sort of, you know, that integrative role that could be played or you could play.

Thanks. I really heard this.

Yes. And Maxim from communication and improvisation.

It's like, yes, there's been a disciplinary way of approaching it and we're going to be working together to come back to first principles or to make it more efficient.

So that's really powerful.

How does active inference help us rethink the nature of online communication where so much of our communication nowadays does occur?

That's that's that's that's a big question, isn't it?

And certainly in the context of social media, politics, fake news and the like, you could you could take that question in lots of lots of different directions,

which I won't do because that's not my field of expertise.

But just off the cuff in terms of first principles and what is communication?

It's the ability for me to infer what you meant.

It's the hermeneutics problem.

If it's the hermeneutics problem, that's most efficiently resolved in terms of dyadic or multi-system interactions.

And when we come back to first principles, which is the generative model, when we share the same narrative or same generative model.

And so in terms of helping, how does it how does active inference help us rethink the nature of online communication?

And I think just from a first principles point of view, it would be the importance of establishing who is talking to who.

And if you want to optimize the efficiency of that exchange, literally, from the point of view of this principle of least action, the speed with which you can resolve uncertainty and minimize your uncertainty or surprise.

And then it's ensuring that like minded communicators are actually communicating because it's only them that will understand each other.

So everybody has to speak the same language.

They have to commit to a shared narrative and a shared generative model.

And then just by things like rate distortion theorem or rewriting that in terms of active inference, the joint free energy minimization between two interlocutors.

And that's the most efficient sort of shared path of least action.

And how does that help engineer or intervene on things?

I'm not so sure.

So just in reference to communications with people, people like Maxwell and the like and other colleagues, there is this interesting notion that if communication, if the real problem of communication is not really the messages that you send, but the inferring whether to send the messages to this person or not, that itself now becomes conditional upon inferring.

So, you know, how does self-organization say, you know, in terms of social media exchange, how is that underwritten by an inference about the kind of people who I am listening to or I am talking to?

And what are the basic principles of that?

Wikipedia page or this news channel?

So understanding how people carve up whether they are like to the degree of similarity to them may be very useful in just getting an idea of the dynamics of message passing amongst communities that will be defined by, on average, how each member of that ensemble individual coarse grains and has a generative model of the kinds of people in the communication grain.

And just to finish this, which is something I've heard, I find a really interesting notion that again would be great if people could simulate this and understand the maths behind it.

Is that the only evolutionary stable from the point of view of the free energy principle, the only one that will be selected by a process of Bayesian model selection.

The only partitioning into in groups and out groups is a 50-50 in group, out group.

In the sense that anything that departs from that sort of dynamically unstable, the evolutionary stable partition means that the smaller group, the out group, the old man out will necessarily ultimately be absorbed into the larger group.

So the only stable partitioning is 50-50, which makes a lot of sense when you look at Trump versus Biden, when you look at Brexit versus not Brexit.

In fact, wherever you look, all the important allegiances in terms of political, ideological, and possibly even theological communication, seems to be split right down the middle.

And perhaps it can be no other way.

So it'd be very, very interesting to simulate that and see if that is a truism that inherits from all of these marginal likelihood or free energy minimizing processes implemented at multiple levels of hierarchical, multiple hierarchical levels, which is, you know, communication is just message passing and message passing is just the way you articulate belief updating and belief updating just is the process of inference which just is the path of least action according to the FEP.

The 50-50 politics, it's maximally confusing, something we all experience.

And a few key points there about the nature of online communication is that at the core, it is dyadic, even when you're broadcasting to many,

it's actually about that connection and the hermeneutic relationship of unpacking meaning.

And then also you brought up the importance of context and identity and who's talking to who and our inferences about that,

which is essential to rhetoric and something that often gets left off when people take big data approaches to online discourse.

The next question is, how does active inference help us think about science communication and participation specifically as we move into broader citizen science initiatives and as scientists are in the loop, something you've been recently involved in as well with society and with decision making.

So as science and the nature of science is changing, who is doing it and how they communicate it, how does active inference help us navigate that?

Right, well, I'm sure you've thought about this much more deeply than I have.

It's just drawing upon my experience, you know, in terms of science communication during the coronavirus epidemic.

I think you're absolutely right.

You can see, as with the previous questions, I think you can take the principles of active inference and just think about what does that mean for optimal communication and belief updating and shared belief updating and shared narratives or not.

And use that as a point of reference for the way that you articulate your own science and you've asked all the challenging and exactly right questions about how you communicate,

how you engage other scientists or other other partners within or beyond academia.

I think the same principles apply exactly to the public.

And just to reinforce your beautiful observation that all communication is dyadic from the point of view of the person communicating.

You know, so it's this kind of person as a unitary object I am talking to or this population or this mentality or this discipline.

So it is, I think, fundamentally dyadic from the point of view of the person generating the messages that may or may not incur belief updating in the recipients.

And these kinds of principles, I'm sure, would be useful in terms of science communication.

So at that level, I don't know that there's much I would have to what you already know and, you know, and we'll all possibly already are implementing.

There is another level, though, which is using active inference, not as a model for the way that we work and communicate and participate,

but as a more statistical or observation model of data.

So in a sense, you can use the principles of active inference really to make the most of data pertinent to a particular domain.

So again, I'm thinking here of the dynamic causal modeling of the epidemiological and behavioral data that has been generated by the coronavirus.

epidemic. So you can certainly use the perceptual inference side of it, if you like, the Bayesian filtering side of it.

But also, in principle, the data mining or the optimal Bayesian design to select which data are useful or not in a very practical way.

when assimilating big data in the service of understanding the system at hand.

So if the system at hand is how does a spike propagate from one neuron to another neuron, the neural network, or how does a virus propagate from one person to another person in the neural population or network,

or a network, your population network, then you can certainly use the data to apply active inference to build generative models of how you think that occurs.

And what immediately comes, you know, you've got to put in all of the things that generate those data.

So you can't miss out any factors that are important, be they psychological, be they behavioral, be they viral, be they transport related.

All of these things have to go into your generative model to best explain the data.

So, you know, when we do this in a practical way, we both use the instance rates from PCR testing and Google mobility data and Department of Transport, transportation data.

Anything that speaks to and reduces uncertainty about all the factors necessary that are entailed by your generative model.

In this instance, it's not a discrete space, it's written down.

In fact, this one is actually a discrete state space model.

And the active inference is not explicitly part of it in the sense that we're not trying to predict people's behavior.

But it does serve an indirect guide through the principles of Bayesian optimum design.

And all that basically means is do I invest computational resources and thereby incur computational and statistical complexity by including or attending to this kind of data or not.

And then you can actually evaluate the information gained by including that data or that data.

So, for example, do you need Google retail estimates and or workplace activity or just one?

If you include both, that means that your complexity increases and you're literally have to wait another half hour before you get the results for your dashboard.

Or do you or do you not have a more parsimonious model in the sense that you have now in the same sense of that saccadic suppression of retinal slip?

And you've actually said, no, I don't need that.

I've got anything I need. I've got the like kind of data just by focusing on these data.

And then once you've got that in mind, you can now go foraging for different kinds of data, different collaborators from different disciplines.

You've got different perspectives, but also crucially different data, you know, to try that will inform and shrink your uncertainty about the model parameters.

And also very importantly, about the structural form of the model.

Do I need this node? How many? Is this interaction important or not?

Is this degree of non-linearity justified by the data?

So all of these questions affected the hypotheses about how this system is responding or would respond if you intervene on it.

All of those questions now become amenable to an evidence based analysis because you've got a generative model underneath the hood.

So that would be a more practical application of active, the principles that underwrite active inference, even though your computer program is not actually doing active inference, but it's certainly been deployed using the principles of active inference.

Awesome. And we heard that integrative approach.

Yes, we're going to include multiple data sets, potentially of unconventional type, and we're going to have a principled way of deciding how to include that data.

And also, as you brought up at the end, who to include in the conversation.

And there was one piece you said in there about the dyadic nature of communication, where a speaker is always, I think you said something like speaking to a person or to a group or to a community.

And it relates to our next question, which is how can we appropriately interact with shared and nested

generative models potentially across scales, be it person, team or community?

Do we think about these levels of analysis as active inference agents in their own right?

Or how do we, for example, speak to a community or speak to a level of analysis that's broader than the personal?

Yeah, I think that's a great and challenging question.

I mean, clearly, there has been some provisional work in academia, looking at sort of Markov blankets of Markov blankets, which is effectively from a stats point of view, from a physics point of view, at least what we're talking about here.

As a physicist, you'll be tackling this with things like the re-normalization or the apparatus of the re-normalization group, which tells you immediately something interesting that the existence of this nested structure, if underwritten by or if it is a re-normalization group, means that there are certain functional forms that are conserved.

So what that means is from a practical point of view, that there will be certain kinds of behavior that are actually conserved at different scales.

So what works in terms of talking to your children should also work as a president talking to your community or a governor talking to your state or a team leader talking to your assembled team.

Simply because in order for there to be a hierarchical nesting that supports that hierarchical structure, there has to be this conservation, usually mathematically written down as a functional form of a Lagrangian.

It could be this sort of marginal likelihood or the surprise that we're talking about that underwrites these sort of paths, most likely paths or paths of least action.

So that actually paradoxically slightly makes the problem slightly simpler.

Well, because what you're saying is what works at one level will work at all levels.

All you've got to do is find the course grading operator operator that takes you from one level to the next.

So what that would look like, I think would be very, very application domain specific.

So, you know, I think that there is a great challenge ahead, which is taking the single particle FEP approach now to into a world that matter where it matters, where the world is actually an ensemble of particles.

And we've already discussed the importance of thinking about worlds where all the particles are identical, whereas all the particles actually vote for half particles vote for Trump and the other half vote for Biden.

And this is interesting to reflect upon pre 21st century physics that was so powerful in articulating this kind of dynamics because it just dealt with the symbol of the

simplifying assumption that my idealized gas was an ensemble of identical particles.

And then you can spin off from that equilibrium physics and everything that led from the, you know, from, from, kind of cycles and, and, and, and engines through to, to, to, to, to, to current, current technology.

So it's a powerful assumption if you just make some simplifying assumptions, but just because, you know, we've already said, well, perhaps that's not the best kind of assumption to make when you're dealing with political mechanics.

You're at least like one by partition in there.

And so that would require a revisiting of the, of that kind of physics, you know, from our point of view or your point of view, basically simulating active agent, inference agents, or ensembles of active inference agents particles.

But where now there's a heterogeneity in play and then asking the questions, well, what are at the next scale, the free energy minimizing or surprising minimizing or potential minimizing solutions at the next scale up.

So we come back to our, you know, why is it the case that people are all split 50 50, which has an enormous impact on the interactions at the scale below.

So I think to tease, I think, or I'm just hand waving here because I don't think there are any, any formal answers.

And I think those formal answers will probably have to come out of agent based and possibly stochastic agent based modeling initiatives.

But with the twist, you're making each agent itself an active inference agent.

So while each individual member of the ensemble is trying to minimize that free energy.

Also the ensemble through cooperation and the shared narrative is minimizing the joint free energy.

And what that means when you move from one scale to the next scale.

You know, this is, if you're in physics, I imagine that this is the problem of beyond non-equilibrium steady states.

Because you're actually now dealing with the multi scale aspect of non-equilibrium.

So at best we have good models of turbulent flow and B-centered actions and laser physics that take us beyond equilibrium physics where all the particles are the same into non-equilibrium physics.

But I don't know that there's an equivalent maths or metaphor in physics that would really speak to the hierarchical nesting.

So I think this is a really open and important research area that I can only recommend is dealt with by numerical analyses basically predicated on underlying principles.

Thanks for the answer there.

And it made me wonder if agree to disagree is a narrative that can be shared even when there is a 50 50 split.

And I think it brings us nicely to the final question of dot coms, which is how can we move people and teams into a co transformative space or as some of your recent work discussed in interactionist space.

Well, I'm sure you know the answers to all these.

I suspect that I'm now realizing you already know the answers because you're learning smiles when I say something that you recognize.

And so actually answering that from on the basis of what you just what you just said.

So I, you know, I think that's another really useful insight that, you know, agreeing to disagree is a surprise minimising,

phase optimal explanation for the exchange with others.

So, but it does rest upon committing to the hypothesis that you are not like me, you are not like minded and that's OK.

So I've now classified you as somebody who's not like minded and I've resolved the ambiguity in among the hypotheses that you are either like minded or you're not like minded.

Normally we resolve those within a few, you know, first impressions within a few seconds based upon all these epistemic cues we offer each other to define the kind of person that we are.

So we make that job as easy as possible or as a signalling to make this so we know our place.

And I use that phrase because of course, you know, there was a paper called Knowing Your Place that exploited a shared generative model that allowed you to be in a particular position in some space, even if you should, if you and I share the same understanding of political ideology, but I know my place because I'm right wing and you're your place because you're left wing or vice versa.

And then we can quite happily exchange, but disagree agreeing to disagree.

So I think that that's a wonderful perspective to have and to endorse it.

And that is a base optimal perspective from both sides of the disagreement.

That's resolving uncertainty in a bounded rational way.

So applying that notion to co-transformative space, it reminds me of the problem where certain patients in psychiatry have committed to a particular inference about whether they belong out there in that kind of environment or not.

So they have decided that they do not belong out there and that those people are not like them and they start to avoid.

So in a very simple, by the way, if there are any psychiatry, there's a whole time in this example.

But I think it's illustrative and useful in that respect.

So take depression or agoraphobia, you know, which is a completely base optimal response.

If I have committed to the hypothesis that out there is full of people who are not like me and potentially will upset, confuse and render me uncertain and possibly even injure me in some way.

So withdraw into that corner of your house or into that silo if you're working in, you know, in teams is a perfectly base optimal response that says that you've got a precise belief that this is where you belong.

This is, you know, these are the people that you speak to and not those people.

So, and that's usually perfectly functional.

In psychiatry, that would be a neurotic defence, but it can become pathological.

And when you become housebound or you become safe, you've got a pathological hypothesis, your body got dysmorphic, you get your you nearly die because of a failure to eat properly.

So when you say co-transformative, I imagine what you mean is you want to transform two teams into one team or at least enable them to work together.

If that's right, and you're not impartially, assume that it is, you're facing the same challenge that a psychiatrist faces in terms of enabling people to revise the precision of their precise beliefs about who they are.

Who they can interact with and who they should interact with.

That's not an easy thing, but it is certainly doable.

And it basically usually reduces to presenting evidence to a group or a person that it can be another way so that they start, you start to revise their prior beliefs or at least the precision of their prior beliefs by in a safe space where it's okay to explore other things.

And enabling them to think about other ways of interacting.

And so this would normally be the objective of psychotherapy basically to by illustration very much in the same way you were talking about sort of, you know, illustrating or educating by embodied experience.

Very much psychotherapy is thought to work like this.

Very much psychotherapy is thought to work like this.

You provide a psychologically embodied experience where you can try out different styles and different

hypotheses.

And in so doing, you paradoxically introduce the right kind of uncertainty about different styles of engagement and who you are and who you are talking to.

And by relaxing that precision, you enable, you give the patient or the or the naughty team that's become too siloed the latitude to explore other ways of behaving.

So I would imagine that most of the tried and trusted procedures to get teams into a co-transformative space and use one or more of those mechanisms.

What would active infants bring to the table?

It would just bring the narrative that everybody trying to get to talk to each other can come to share so they can see through the process of becoming more collaborative or exchanging ideas more fluently.

Or working to working with the same lexicon or mechanics or code.

And it will enable them.

It will enable them to.

Having the same narrative will actually shape their prior space and understand the mechanics of actually enlarging the hypothesis space in terms of interaction styles.

So that was an incredibly hand waving answer.

But it was in part informed by almost my understanding of the question from the point of view of a psychiatrist who wants to transform the way that a patient relates to her world.

And drugs can help.

I mean that literally.

Those drugs that are responsible neurobiologically for setting the precision.

If you can temporarily suspend the precision in order to reveal other latent a priori hypotheses in terms of the way that I am or the way that I interact and the way that I behave or the way that I perceive.

That can actually have long lasting effects on bringing those other hypotheses to the table in the moment in subsequent interactions.

So perhaps the most compelling example of this, which is trending at the moment, is the use of psilocybin assisted therapy, particularly in terminal care.

So, you know, if you know you're going to die of cancer in the next six months, there are certain hypotheses that are brought to bear in terms of how I would expect to feel and how I engage with the world and how I engage with my loved ones as a dying person who is near death and the ultimate loss.

Those are not necessarily the best or most functional hypotheses or ways of being.

There are other ways of dying gracefully and gloriously, but to get at them, sometimes having a managed challenge to your 5-HT_{2A} receptors via things like psilocybin and other related drugs just allows you to suspend for a moment your very precise beliefs about the kind of thing that you're going to be.

It allows you to experience other ways of being and perceiving, which can be very useful when it comes to just trying out other hypotheses in your, in your, in this instance, your cancer journey.

But, you know, we can also imagine similar scenarios when you get locked in to a particular way of interacting with either within a team or between teams in a larger organization.

So that would mean you have to go on a retreat and take lots of magic mushrooms.

Never thought I would hear it from you, Professor Friston, but there we have it. Drugs for Teams.

Thank you for this excellent interval with Dotcom's unit.

We're going to take another five-minute break and we'll return for the final session for DotTools.

Thanks again, everyone, and we'll see you in five minutes.

That will be five minutes.